

STUDY GUIDE

UNITED NATIONS ENVIRONMENT ASSEMBLY



Agenda 01: Deliberating on the Environmental Implications of Resource Extraction in Polar Regions

Agenda 02: Evaluating the Impact of Climate-Altering Technologies and Measures





Letter from R&D

Dear Delegates,

It gives the Department of Research & Development, MUN Society MPSTME, immense pleasure to welcome you to the 12th edition of Mumbai MUN! True to our legacy, we remain committed to upholding and building upon a tradition of fostering informed discussions on matters of widespread concern.

We are primarily responsible for formulating relevant agendas and developing study guides that aim at imparting a holistic understanding of the issues at hand. This document has been prepared with a *first principles* approach, meant to play a part in orienting the committee discussions towards one that rises above the convention, and is characterised by substantiveness and inclusivity.

This year's theme, 'Voice Amidst the Void,' stands to remind us that when ideals fade into absence, voices must prevail. In an increasingly complex world, with a lack of conviction and shared purpose, we believe that the introduction of new ideas and impactful discourse is the need of the hour. True progress requires us to have the courage and willingness to speak up, have our perspectives heard, and to make sure that no voice is lost amidst a void of apathy.

Keeping this theme in mind, we have deliberately curated committees and agendas that focus on issues often overlooked in global conversations to foster meaningful dialogue and resolution where it is most needed. In this study guide, you will find well-researched background information to help you prepare for your committee sessions. As facilitators, we aim to equip you with the resources and insights you need to make the most out of your experience at Mumbai MUN.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference an inspiring, enriching, and especially memorable experience for all.

Best wishes,
Department of Research & Development,
MUNSociety MPSTME.

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Agenda 1 - Deliberating on the Environmental Implications of Resource Extraction in Polar Regions

COMMITTEE OVERVIEW

The United Nations Environment Assembly (UNEA) is a committee for developing policies for the environment. It has a universal membership of 193 states. It provides policy guidelines, undertakes policy review and sets strategic guidance regarding environmental challenges. It also outlines a direction for the United Nations Environment Programme (UNEP) and creates opportunities for partnerships to reach cumulative goals set for ecological progress.

In 2012, a document was created called 'The Future We Want'. Subparagraph 88 (a) of this document provides for universal membership in the governing council of UNEP. Hence, the fifty-four-member governing council was renamed as the United Nations Environment Assembly, consisting of all 193 members. It was then decided that the Environment Assembly would establish the global environmental agenda. The UNEA Bureau includes one president and eight vice-presidents. These members, along with a rapporteur, are elected at the final meeting of a session.

AGENDA OVERVIEW

The polar regions are currently facing several issues, from global warming to the threats posed to the native population. Rising resource extraction aids in exacerbating these, and poses an issue in itself. The agenda focuses on these challenges and their impacts on the

environment. Furthermore, discussions will also revolve around the technical and historical aspects of extraction in polar regions, which will then lead us to our actions now. The agenda describes the grave situation of the environment in the polar regions due to the extraction of resources. This is inclusive of both current and future statistical prediction status of the environment.

BACKGROUND

Due to the historical and global interest of the Antarctic continent, as well as the emotional investment of several cultures, upon its discovery, it was declared to be an international natural sanctuary, with no single entity to lay claims over it.¹ This ruled out the possibility of Antarctic exploitation on a large scale.

On the other hand, as more parts of the Arctic become accessible and navigable by sea, trade routes and resource-rich regions only increase the importance of Arctic governance, and contenders rise to reap the economic benefits of an environmental mishap.²

In the early modern era, the US and Canada utilised ice-strengthened drill ships and temporary gravity-based platforms to facilitate underwater drilling in shallow regions, but only during the summer months. Other ingenious methods include building a spray ice island. In the summer of 1987, seawater was pumped onto floating wooden platforms, which froze in the Arctic climate, creating floating frozen island-like platforms to perform their mining operations.³ As

¹ (PDF) *Cultural Connections with Antarctica and the Southern Ocean*. (n.d.).

https://www.researchgate.net/publication/390616604_Cultural_Connections_with_Antarctica_and_the_Southern_Ocean

² Team, P. I. C. (2025, April 4). *Arctic Region: Significance & Geopolitical Tensions*. PMFIAS.

<https://www.pmfias.com/arctic-geopolitical-tension/>

³ Noble, P., & Maddock, B. (2015). *Historical background on Arctic exploration and production technology development*. In *Working Document of the NPC Study: Arctic Potential: Realizing the Promise of U.S. Arctic Oil and Gas Resources* (p. 1) [Report].

<https://www.npcarcticreport.org/pdf/tp/6->

[2_Historical_Background_on_Arctic_Exploration_and_Production_Technology_Development.pdf](https://www.npcarcticreport.org/pdf/tp/6-2_Historical_Background_on_Arctic_Exploration_and_Production_Technology_Development.pdf)

time went by, more sophisticated methods became the industry standards, and now, year-long permanent mining platforms, e.g., Pirazlomnaya in the Barents Sea, are commonly observed.

History of Extraction in Polar Regions

Early European colonisation

In the late 16th century, European nations such as England, the Dutch Republic, and Norway and Sweden from Scandinavia set sail to the previously undocumented parts of the globe, which they referred to as the “Arctic”, a barren, icy region, sans vegetation.⁴

Gold Rush and Early Mining

After the deeply criticised sale of Alaska to the USA, several expeditions began to evaluate the economic feasibility of this decision. This scrutiny resulted in the discovery of gold deposits such as Kensington and the Brooks Range in the southeastern provinces of Alaska. This turned into a gold rush similar to that of the mid-1800s Californian gold rush and resulted in large-scale migration and the formation of settlements, trade routes, and mining infrastructure in the new continent.⁵

A similar reaction was observed in the Canadian populace, which was attracted towards the Yukon province, feeding into the Canadian gold rush. This resulted in rare earth minerals and gold being found in the new continent. Meanwhile, in Europe, the Scandinavian countries began to

double down on their coal-rich cut of the Arctic. The Svalbard coal mine from Norway was the largest and most productive mine among all the Scandinavian nations, which directly fuelled the flames of the Industrial Revolution.⁶

Oil and Gas Development

In the mid-20th century, the global focus shifted to hydrocarbons and petroleum extraction. Arctic surveys found large deposits of offshore oil and natural gas reserves concerning various areas of the Arctic Circle, including the Amerasian Basin, the East Barents Basin, the East Greenland Rift Basins, and the West Siberian Basin.⁷

With the discovery of oil and natural gas in this harsh and barren Arctic ecosystem, the extraction of oil became a novel challenge. This required the development of new ingenious technologies such as frigid oil rigs, newer marine vessels known as “icebreakers”, polar pipelines, etc. These efforts extensively helped the Arctic extraction market to grow and prosper.

Areas of Extraction

Coal and diamond mines - Norway and Russia both operate the last two coal mines on Svalbard, while Canada mines diamonds in the resource-rich northwestern provinces.⁸

Minerals - Cryolite, given by the chemical formula Na_3AlF_6 , is a crucial component in aluminium enrichment. Before closing in 1987, Greenland's

⁴ Secret Atlas. (2025, September 23). The history of Arctic Exploration | From Barentsz to the North Pole. <https://www.secretatlas.com/handbook/culture-and-history/arctic/the-history-of-arctic-exploration>

⁵ The Arctic: a history. (2017, July 5). New Internationalist. <https://newint.org/features/2009/07/01/arctic-history>

⁶ Britannica Editors. (2025, September 20). Klondike gold rush | Yukon Territory, Prospectors, Discovery. Encyclopedia Britannica

<https://www.britannica.com/event/Klondike-gold-rush>

⁷ Oil survey says Arctic has riches. (2008, October 1). AAPG. <https://www.aapg.org/news-and-media/details/explorer/articleid/47955/oil-survey-says-arctic-has-riches>

⁸ Jones, F. (2023, April 19). “A ‘frozen’ geopolitical issue”: Svalbard and the geopolitics of Arctic mining. Mining Technology. <https://www.mining-technology.com/features/arctic-mining-svalbard-russia/>

Ivittuut mine was the world's largest cryolite deposit.⁹

Metals - Russia exploits large reserves of nickel and palladium from Norilsk, the northern part of the Arctic Circle, and gold in Northern Siberia.¹⁰

Oil and Gas - 30% of the world's undiscovered natural gas and 13% of the world's oil reserves reside in the Arctic.¹¹ The prominent quantities of these hydrocarbons are found in the Barents Basin near Norway and Russia, Arctic Alaska, as well as the Amerasian Basin, which is found near Greenland, Canada, and Alaska.¹²

GEOPOLITICS IN THE REGION

Russia vs NATO

The Arctic is important in terms of resources. In recent years, Russia has been remilitarizing the Arctic, reopening the old Soviet-era research and military bases, and increasing its intelligence presence through new radar technology. This is seen as a response to increasing NATO presence in the Arctic through Scandinavian countries like Finland and Sweden, which recently gained NATO membership, threatening the Arctic co-operation that is being practised among the

involved countries, especially between Russia and the collective West.¹³

Greenland Purchase Proposal

The US administration has, in recent times, expressed interest in purchasing Greenland, an autonomous territory under Denmark. This is owing to the fact that the island is rich in natural resources and provides unrestricted access to the North Atlantic sea route. The issue was first sparked by a statement made by the American president, Donald Trump, in 2019, stating that the USA is considering all options to obtain Greenland.¹⁴ These statements have regularly been criticised and responded to by both the Greenlandic and the central Danish authorities. These events have contributed to increased tensions in the Arctic region and jeopardised the notion of long-term regional stability.¹⁵

Northwest Passage Dispute

The Northwest Passage is the sea route connecting the Atlantic Ocean to the Pacific Ocean through the Canadian Arctic Archipelago. This is the point of disagreement between Canada and the US that makes the American cut of the Arctic devoid of agreement. Canada claims this northwestern passage as internal waters,

⁹ North of 60 Mining News. (2025, August 1). Cryolite: Greenland's forgotten icy mineral.

<https://www.miningnewsnorth.com/story/2025/08/01/mining-history/cryolite-greenlands-forgotten-icy-mineral/9197.html>

¹⁰ Pistilli, M. (2025, June 9). Discover the top 9 Nickel-Producing countries shaping the global market. Investing News Network (INN). <https://investingnews.com/daily/resource-investing/base-metals-investing/nickel-investing/top-nickel-producing-countries/>

¹¹ Arctic oil and natural gas resources - U.S. Energy Information Administration (EIA). (n.d.). <https://www.eia.gov/todayinenergy/detail.php?id=4650>

¹² NPC. (n.d.-a).

NPC. (n.d.-a). https://www.npc.org/AR-2015-3-27_NPC-Council_Meeting-FINAL.pdf

¹³ Boulègue, M., Ålander, M., Collén, C., Luc,realizinga, E., Sendak, C., & Viksnins, K. (2024, December 5). Up North: Confronting Arctic insecurity implications for the United States and NATO. CEPA. <https://cepa.org/comprehensive-reports/up-north-confronting-arctic-insecurity-implications-for-the-united-states-and-nato/> because

¹⁴ Trump sets sights on Canada, Greenland and the Panama Canal. (2025, January 8). [Video]. NBC News. <https://www.nbcnews.com/politics/donald-trump/trump-suggests-use-military-force-acquire-panama-canal-greenland-econo-rcna186610>

¹⁵ Khalil, H. (2025, April 3). Denmark and Greenland show united front against US "annexation" threats. <https://www.bbc.com/news/articles/cr4nlgwzy6qo>

along with the sovereignty and decision-making power that comes with it. The United States has made the argument that passage comes under UNCLOS (United Nations Convention on the Law of the Sea), providing all ships travelling in international waters with innocent passage.¹⁶

Svalbard Dispute (Norway vs Russia)

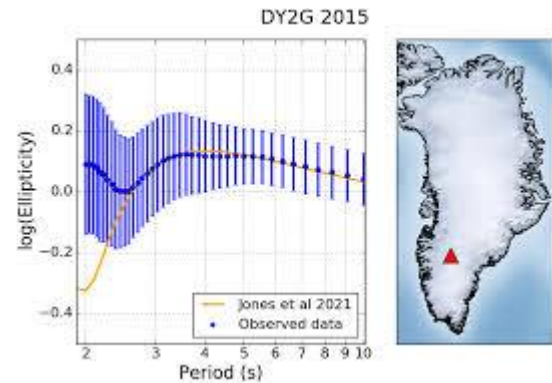
In 1920, the Svalbard Treaty was signed primarily between the European powers, such as the UK, Russia, France, Italy, and the Netherlands, and Scandinavian countries like Norway and Sweden. This treaty stated equal rights for the signing nations to exploit the region for its economic use while recognising absolute Norwegian sovereignty over the Svalbard archipelago.¹⁷ Under the following treaty, Russia still operates two mines in the Svalbard region and conducts extensive fishing. They have also reopened a research base with the collaboration of the BRICS nations.

METHODS OF EXTRACTION

Rayleigh Wave Ellipticity

Seismic waves that travel along the surface of the earth in a receding motion (elliptical), causing particle displacement, are called Rayleigh waves. The wave frequency defines the sensitivity for the elliptical measurements, and they are directly proportional. Advancements in the research have made it possible to track ice formation and disintegration. The Earth has natural sounds that are consistent throughout the globe. These include seismic noises that have 2 peaks, namely, Primary Microseisms(10-20s) and Secondary Microseisms(1-10s). Both of which are the result

of the interaction of gravity waves and the rock bottom of the ocean, producing seismic waves.¹⁸



Left: ellipticity measurements. Right: Location of DY2G in Greenland (red triangle)

Source: Extraction and applications of Rayleigh wave ellipticity in polar regions, 2023

Another application of Rayleigh waves is the characterisation of Subglacial Geology. The characterisation is done by Earth's constant hum. High wave ellipticity is extracted from this hum, which has proven to be sensitive to the superficial structure of the crust. The Rayleigh wave ellipticity is thus useful in measuring the ice thickness and imparting knowledge of the internal structure of the ice.

Solid Liquid Extraction (SLE)

It is the disintegration of the solid component into the liquid solvent. It provides a stable and undisturbed platform for interaction with the liquid phase. The entire process is based on pouring the fluid onto the SLE cartridge for the solid surface to absorb. An immiscible solvent is then added on top, and an amalgamation of undesirable salts and polar interferences is left behind. These products help in the quick and accurate preparation of samples, since there is

¹⁶ Tømmerbakke, S. G. (2019, April 12). 30 Year-Old compromise divides USA and Canada. *High North News*.

<https://www.highnorthnews.com/en/30-year-old-compromise-divides-usa-and-canada>

¹⁷ Jones, F. (2023, April 19). "A 'frozen' geopolitical issue": Svalbard and the geopolitics of Arctic mining. *Mining Technology*. <https://www.mining-technology.com/features/arctic-mining-svalbard-russia/>

¹⁸ Extraction and applications of Rayleigh Wave Ellipticity in polar ... (n.d.).

<https://discovery.ucl.ac.uk/id/eprint/10174255/1/extraction-and-applications-of-rayleigh-wave-ellipticity-in-polar-regions.pdf>

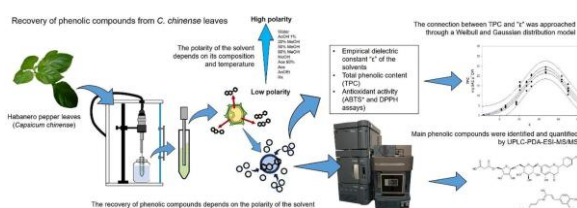
one solid component and one liquid; no mixtures are formed. It is an automated, time-preserving method of extraction. SLE is a rapid extraction method that helps in sampling biological, food, and environmental samples.¹⁹

Microwave Assisted Extraction (MAE)

It is based on separating emulsions of compounds with distinctive solubility. The analysis takes place after phase separation when the particular particles of interest are transferred into the solvent. This method is used to extract organic compounds from various origins. The fusion of microwave energy and effective magnetic stirring has proved to be extremely beneficial in terms of speed and solvent consumption. It helps in the preservation of compounds and increases the yield of this extraction. Along with multifunctional microwave ovens, specific function-based reactors have also become relevant.²⁰

Ultrasound Assisted Extraction (UAE)

It supports cell wall degradation and the transfer of a significant amount of solute particles to mix with the solvent. This helps in the successful inhibition of phenolic compounds to boost TPC recovery.



Graphical abstract of Ultrasound-Assisted Extraction
Source: Effect of solvent polarity on the Ultrasound Assisted extraction, 2021

The phenolic recovery is paired with antioxidant activity, which is greatly impacted by the polarity of the solvent.²¹ UAE employs a sound frequency of nearly 20kHz, which is produced by converting electrical energy into mechanical energy that has the ability to travel through fluids.²² UAE is applied at an industrial level since it offers great speed, sustainability and produces greater yields than conventional methods.

DEVELOPMENT OF TECHNOLOGY

Ocean Glider Development Trajectory

In 1989, Henry Stommel proposed the idea of a massive robot array that would have a certain longevity. It would also be capable of collecting and providing insights into measurements of vast areas of the ocean. At the same time, Craig Lee was the one who had commenced this discussion of the development of ocean glider technology. Lee then developed an outline for the timeline for this development to occur, which was as follows: the first 5 years would be initial prototyping, the next 10 years for application, and another 15 years to spread and adapt this technology. The numerous challenges during the development of this technology include recruiting skilled people to use the technology, low-scale commercial production due to the specialised hardware, as well as logistics, including shipping, transport, and deploying the equipment. These issues continue to persist, and scientists are trying to

¹⁹ SLE. Agilent. (n.d.). <https://www.agilent.com/en/product/sample-preparation/supported-liquid-extraction-sle>

²⁰ Microwave-assisted extraction: Anton Paar Wiki. Anton Paar. (n.d.). <https://wiki.anton-paar.com/in-en/microwave-assisted-extraction/>

²¹ Effect of solvent polarity on the ultrasound assisted extraction and antioxidant activity of phenolic compounds from habanero pepper leaves (capsicum chinense) and its identification by UPLC-PDA-ESI-MS/MS - Sciencedirect. (n.d.-a). <https://www.sciencedirect.com/science/article/pii/S1350417721002005>

²² Kinetic ultrasound-assisted extraction as a sustainable approach for the recovery of phenolics accumulated through UVA treatment in strawberry by-products. Foods (Basel, Switzerland). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10453509/>

find solutions with the assistance of research and funding.²³

Drilling Technologies

There are several scientific communities that rely on drilling technologies to uncover what is buried beneath the ice. A major concern was that this could cause disputes between the efforts to share the research and their extraction sites. Another significant issue raised by the participants was that these drilling operations could lead to the discovery of an organism that could possibly be hazardous to our environment.

FUTURE PROSPECTS

AutoSub

Autosub vehicles provide 2 variations: The AutoSub long-range vehicles and the AutoSub 5. They have employed studies regarding under-ice operations powered by these vehicles. AutoSub 5 is the latest model, which is designated for under-ice operations. They built two propeller systems to yield redundancy. There have been several discussions on the Guidance, Navigation, and Control (GNC) system. The arrival of recent issues, such as harsh underwater conditions and managing to keep the costs low while providing autonomous marine platforms, has created the need for solutions using the GNC system. This modular part controls flexibility and helps in communication between various parts of the vehicle. The possibility of combining these features and supporting compatibility during marine operations inspired the desire to create control systems for these devices. These could be adapted to several autonomous robotic

platforms and have various characteristics like vast actuation, sensing, and hydrodynamics.

Augmenting GNSS sites

The Ohio State University had a discussion on a few topics regarding the improvement in the GNS system, including the development of a global navigation satellite system (GNSS). The Resolute Polar Receiver caused GNSS to transform and was thus integrated with the Iridium satellite network, which led to a major decrease in the power demands. Augmenting GNSS sites with absolute gravity sensors is a significant example of accepting an extensive array of instruments based on communications and power. This is used to calculate measurements to resolve any discrepancies in ice-mass change. A new possibility is being explored, of putting geophysics sensors on a new platform, which will later be dropped from helicopters to a few inaccessible regions of the polar region. This is currently a working system in Australia, but it has not been applied to the polar regions. This idea is derived from Geoscience Australia's Onshore Energy Security Programme (OESP).

CHALLENGES

Warming of Polar Regions

Global climate change has a massive impact on the polar regions. According to statistics, the Arctic is heating 7 times faster than the mid-latitudes. One of the main reasons for air pollution in the region is due to gas flaring. It accounts for 42% of the Arctic black carbon emissions.²⁴ The warming of this region, thus,

²³ *Technology developments to advance Antarctic Research: Proceedings of a workshop at nap.edu*. 3 Technologies for Research and Observational Instrumentation | Technology Developments to Advance Antarctic Research: Proceedings of a Workshop | The National Academies Press. (n.d.).
<https://nap.nationalacademies.org/read/26699/chapter/4#10>

²⁴ Ekaterina Borshchevskaia, V. G. (2025, July 19). *Pollution in the arctic: Oil and gas extraction of the Continental Shelf as a major contributor*. The Arctic Institute - Center for Circumpolar Security Studies.
<https://www.thearcticinstitute.org/pollution-arctic-oil-gas-extraction-continental-shelf-major-contributor/>

demands changes in the lifestyle of the species and at least 5 million people. The speed of revival of local plant communities in the Arctic is relatively much lower than in the south. In 1987, the USA joined in to provide an incentive to offshore oil yields. As it turns out, Russia has a lot of control over the polar region and thus 90% of it is contributed by gas and the other 10% by oil. The oil and gas extraction taking place in this Arctic region has caused national energy security concerns for both Russia and the USA.

Oil Spills and Oil Mixing

Due to the constant transport of oil and petroleum, land is damaged. To reclaim some of this land, sand is sprinkled around the oil, so it mixes with the sand and further seeps into groundwater and reservoirs and finally ends up in the Arctic Ocean. Thus, this increases the pollutants in regions like the Barents, White, and Kara. Oil spills are capable of erasing a generation of flora and fauna, and they could also deeply affect the native people.²⁵ The liquidation of oil becomes tricky in polar regions where the temperature drops so low, and there is no way of harnessing this oil once it gets underneath the ice sheets. Moreover, since the Arctic is in a remote location and there is not enough infrastructure to tackle an oil spill, by the time treatment is done or aid has arrived, pretty severe damage is done. An oil spill near Norilsk, above the Russian Arctic Circle, for instance, entered a glacial lake. The spill occurred on 29th July, 2025, and there was a

leak of around 20,000 tons of diesel. The local officials were acquainted with this news days after the spillage which led to Russian President Vladimir Putin declaring a state of emergency.²⁶

Lack of Development of New Technology

While older drilling technologies and other extraction technologies, such as distillation, are still in use, these methods are difficult to manage in huge batches of resources and are thus not feasible. In recent times, researchers have come up with MAE and UAE, but the development of these methods was a tedious and long process. While strategies such as GNSS are still in the development stages. Other drawbacks are the heavy pricing of MAE due to the equipment needed to run the process.²⁷ This also makes their market sales challenging.

EFFECTS

Increase in Damaged Land

Oil and petroleum consistently pollute our environment since they are mobile resources. The area of damaged land increases by 10,000 hectares every year. According to the statistics, for every 100km of pipelines, 500 hectares of land are damaged.²⁸ The existing technologies to reverse the damage to the land are ineffective. However, there are a few technologies to help in remediation of the damage caused, namely Permeable Reactive Barriers (PRB), chemical fixation, and phytoremediation. These methods

²⁵ Ekaterina Borshchevskaia, V. G. (2025, October 10). Pollution in the arctic: Oil and gas extraction on the Continental Shelf as a major contributor. The Arctic Institute - Center for Circumpolar Security Studies.

<https://www.thearcticinstitute.org/pollution-arctic-oil-gas-extraction-continental-shelf-major-contributor/>

²⁶ Diledhad, gges, C. (2020, June 9). As Arctic oil spill spreads, Bellona offers assistance. Bellona.org.

<https://bellona.org/news/arctic/2020-06-as-arctic-oil-spill-spreads-bellona-offers-assistance>

²⁷ Technology developments to advance Antarctic Research: Proceedings of a workshop at nap.edu. 3 Technologies for Research and Observational Instrumentation | Technology Developments to Advance Antarctic Research: Proceedings of a Workshop | The National Academies Press. (n.d.).

<https://nap.nationalacademies.org/read/26699/chapter/4#10>

²⁸ Ekaterina Borshchevskaia, V. G. (2025, October 10). Pollution in the arctic: Oil and gas extraction on the Continental Shelf as a major contributor. The Arctic Institute - Center for Circumpolar Security Studies.

<https://www.thearcticinstitute.org/pollution-arctic-oil-gas-extraction-continental-shelf-major-contributor/>

have different capabilities of handling multi-metal contaminated sites.²⁹ This reversal is only processed when the land is near a road; the regions outside the roadside section are not reclaimed. More fields are being developed to extract oil and gas, which will only add to the damaged land area and will cause an increase in pollution of the environment.

Damage and Disturbance to the Native Population

The existing population of the polar regions already faces challenges related to weather and sourcing food. Extraction carried out in these regions only exacerbates their problems. The opening of several mining sites has displaced them and caused them to move to a different region. The polar regions consist of a lot of infrastructural and industrial development in their future; thus, a balance must be found between providing employment and preventing any harm to the environment. The increased pollutants in the air and water cause difficulty for different species to continue to exist in that region, and medical help is not easy to come by in such regions.

PAST ACTIONS

Previous UN resolutions

United Nations Convention on the Law of the Sea (UNCLOS)

UNCLOS is an international treaty under the purview of the United Nations. It provided a framework that was used to ratify maritime control over countries' coastal waters, clearly

setting boundaries between that and the international waters. This treaty created several new international bodies, such as the International Tribunal for the Law of the Sea (ITLOS), where international maritime disputes are brought forth with the intent to resolve and start dialogue between the disputing parties. The International Seabed Authority (ISA) is the UN body that establishes the commission to internationally recognise whether a seabed is a continuous extension to a nation's subcontinent, providing them with the rights to extract those resources from the seabed.³⁰ It also deals with land reclamation laws concerning the continental shelf and artificial islands. UNCLOS also regulates the right to marine resource exploitation and species-biodiversity manipulation.

Article 234 - UNCLOS

Under UNCLOS, in 1982, Article 234 was specifically composed for the Arctic waters, providing countries with a special status and the power to limit navigational authority in order to preserve the fragile Arctic ecosystem and prevent marine pollution in their EEZs.³¹ Under the provisions of this article, countries like Canada and Russia have implemented stricter shipping rules, from decreasing the vessel discharge and ship dimensions to travelling in the Arctic waters. This access and control do conflict with the fundamental right to freedom of navigation under UNCLOS itself.

Protection of Arctic Marine Environment (PAME)

PAME, a part of the Arctic Council, comprises the Arctic 8 countries in addition to some indigenous

²⁹ Camenzuli, Danielle, et al. "On-Site and in Situ Remediation Technologies Applicable to Metal-Contaminated Sites in Antarctica and the Arctic: A Review." *Polar Research*, vol. 32, no. 1, 31 Dec. 2013, pp. 1–15, www.tandfonline.com/doi/full/10.3402/polar.v32.21522#d1e237

³⁰ Garg, R. (2022, August 28). International Tribunal for the Law of the Sea (ITLOS) - IPLeders. iPleaders. <https://blog.ipleaders.in/international-tribunal-for-the-law-of-the-sea-itlos/>

³¹ Solski, J. J. (2021). The genesis of Article 234 of the UNCLOS. *Ocean Development & International Law*, 52(1), 1–19. <https://doi.org/10.1080/00908320.2020.1835026>

observers. They create strategic plans and guidelines to counter Arctic pollution.³² In addition to the UNCLOS, these nations come together in this forum as a united body, with a united stance of ensuring that the Arctic biome is preserved. Though it does not pass any internationally enforceable laws, its guidelines are followed due to the political weight that is being carried by the collective eight.

Important Treaties

High Seas Treaty

Ratified in 2023, the High Seas Treaty, also known as Biodiversity Beyond National Jurisdiction (BBNJ), was a landmark treaty that explained and compiled governance policies for marine biodiversity and preservation of the Arctic ecosystem outside their respective Exclusive Economic Zones. It tries to address conflicts of interest in Arctic decision-making, with the central aim of protecting and safely developing the Arctic. All Arctic coastal nations except for Russia have signed the High Seas Treaty.

Antarctic Treaty

Signed in 1959, the Antarctic Treaty has been the foundation for Antarctic governance, which banned all resource extraction activities on Antarctic soil, except for research purposes.³³ The Madrid Protocol, signed in 1991, designated Antarctica as a "natural reserve devoted to peace and science."³⁴ It was the result of several decades' worth of protests and dialogues, which led to a moratorium on mining in 1976, all the way up to the Madrid Protocol. Hence, unlike the Arctic, the Antarctic has been devoid of exploitation incurred as a result of mining and extraction.

CASE STUDIES

Effect of Polar & Non-Polar Solvents

A 2023 study comparing the efficiency of flaxseed oil extraction methods using polar and non-polar solvents confirmed that non-polar solvents showed a higher yield in seed oil extraction. What it means is that, when using non-polar solvents like hexane and petroleum ether, the effective oil yield observed from flax seeds was 40%. In the case of polar solvents such as ethanol and methanol, however, the effective oil yield observed was 35%.³⁵ Though the lower yield resulted in a lesser quantity of oil extracted, the active biomolecules present in polar solvents improve the overall nutrition profile of the extract, increasing its quality for consumption.

This suggests that different extraction methods should be implemented for differing scenarios. In case of large-scale industrial functioning, non-polar solvent extraction techniques are used to maximise oil output, and while producing oil for edible purposes, polar solvent extraction techniques are used to maximise the nutritional content of the flaxseed oil.

Climate Change Impacts and Resilience: An Arctic Case Study

Climate change refers to the alteration in the climate over a period of time. The anthropocene process has contributed largely to Greenhouse gas (GHG) emissions. Human activities have caused about a 1 °C change in the global temperature. The snow cover in the northern hemisphere of the Earth has decreased due to the rise in the melting of ice sheets. Efforts have been

³² Protection of the Arctic Marine Environment (PAME). (n.d.). UNEP - UN Environment Programme. <https://www.unep.org/protection-arctic-marine-environment-pame>

³³ The Antarctic Treaty | Antarctic Treaty. (n.d.). <https://www.ats.aq/e/antarctic treaty.html>

³⁴ Environmental Protocol | Antarctic Treaty. (n.d.). <https://www.ats.aq/e/protocol.html>

³⁵ Ekka, S., & Oway, J. (2023). Extraction effect of polar and Non-Polar solvents on the physio chemical characteristics of flaxseed oil. Biosciences Biotechnology Research Asia, 20(4), 1511–1517. <https://doi.org/10.13005/bbra/3196>

made to reduce emissions, yet GHG emissions continue to damage our environment.³⁶

Climate change is not dependent on the region; GHG emissions can cause harm throughout the globe. Communities must become more resilient towards these harsh conditions so that we can collectively adapt and evolve. There is not enough research on how climate change in a given region affects particular communities, and this leads to a lack of knowledge, and thus, local adaptation policies are not created, which makes the process of assembling resources tricky.

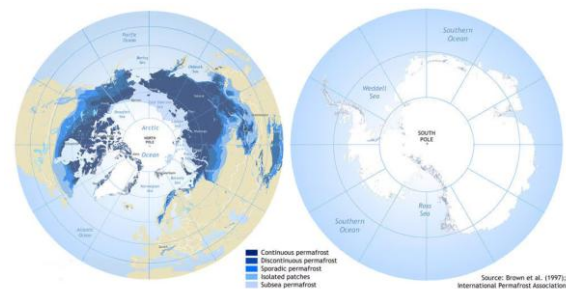
Local governments often deal with issues that are time sensitive and allocate resources according to short-term issues, since future problems are conflicting, and arranging resources for future issues is much more overwhelming.

The hierarchy of every government should coordinate and integrate its adaptation policies to unite the nation/state; this is another way of building resilience.

Arctic Permafrost Impacts

The first and only oil well was created in 1986 in the Arctic National Wildlife Refuge, and since then, several oil wells have been formed. The thaw rates threaten the infrastructure and roads; they also create a marsh cover on the snow, which makes transport more difficult. Each and every single oil well leaves a mark on that specific land, which in turn causes the permafrost underneath to change. An experiment was held to integrate the data of long-term impacts with short-term impacts. 4 gravel pads were placed, and a 100m radius was taken to measure the impact.

The comparison of old and new oil pads tells us the primary and secondary effects of the oil and gas industry on the permafrost tundra. This study imparts knowledge on how the addition of an oil pad changes the surroundings. The temperature increases near the location of the pads. These four pads are much smaller than the huge industrial oil pads, and they are capable of providing knowledge of the activity above ground.³⁷



Effect of permafrost thawing on carbon emissions across the polar region.

Source: A genomic analysis using RNA-seq (2012).

CONCLUSION

There have been significant changes to the environment since resource extraction began in the polar regions. From climate change to the displacement of the native population, there have been direct impacts on the species currently inhabiting the polar regions. Nonetheless, there have been developments in technology to ease this process of extraction. This agenda helps us in weighing the benefits of polar extraction against the demerits. It is important to note that while nature offers many resources, these are finite, and their exploitation can lead to several ill-effects. Hence, deliberation on resource extraction in the polar regions becomes necessary.

³⁶ (PDF) climate change impacts and resilience: An arctic case study. (n.d.-b).

https://www.researchgate.net/publication/332773358_Climate_Change_Impacts_and_Resilience_An_Arctic_Case_Study

³⁷ Miner, K., Baskaran, L., Gay, B., Sousa, D., & Miller, C. (2024, October 25). Frozen no more, a case study of Arctic permafrost impacts of oil and gas withdrawal. Nature News. <https://www.nature.com/articles/s41598-024-76292-2>

Agenda 2 - Evaluating the Impact of Climate-Altering Technologies and Measures

AGENDA OVERVIEW

This agenda aims to understand climate-altering technologies and evaluate their effectiveness, side effects, and global implications. These technologies range from a wide spectrum of conventionally used methods like afforestation to newer radical methods like cloud seeding. The technologies mentioned in this paper have the potential to slow climate change and possibly even reverse it, but carry severe side effects, uncertainty, and international tension. This agenda stresses the importance of designing policies that ensure climate solutions are safe, fair, and responsible.

INTRODUCTION

Climate-altering Technologies and Measures (CATM), also sometimes referred to as climate or geo-engineering, refers to a broad set of methods and technologies that aim to deliberately alter the climate system in order to alleviate the impacts of climate change. This is done in order to stay on track to achieve the 1.5°C targets, if not 2°C. The International Panel for Climate Change (IPCC) has stressed that there is no other alternative but to employ large-scale CAT to achieve global temperature goals.³⁸ This could be achieved with CATs through two profound techniques, namely Solar Radiation Management (SRM) and Carbon Dioxide Removal (CDR). All CATs could be seen as some

derivative or combination of these two technologies.

BACKGROUND

Modern historical developments include:

1. 2009: Geoengineering the Climate, published by The Royal Society, formally introduced the idea of geoengineering for climate change on the global stage.³⁹
2. 2012: CarbFix, founded in 2007, started its large-scale pilot project of injecting CO₂ into Iceland's basalt rock in order to mineralise it; another large-scale initiative was launched by Canada under the name 'Haida Project' later that same year, in July, where large amounts of iron supplements were strategically dumped in the ocean to stimulate exponential plankton growth.
3. 2013-14: IPCC's 5th summit stressed active involvement of CATs to fulfil the 1.5-2°C goals.⁴⁰
4. 2019: UNEA-4 (United Nations Environment Assembly), through Switzerland, officially drafted a resolution for "Geoengineering and its governance".⁴¹ Though this resolution was withdrawn, it marked the beginning

³⁸ United Nations. (n.d.). 1.5°C: what it means and why it matters | United Nations.

<https://www.un.org/en/climatechange/science/climate-issues/degrees-matter>

³⁹ Geoengineering the climate: Science, governance and uncertainty. In RS Policy Document 10/09.

<https://royalsociety.org/-/media/policy/publications/2009/8693.pdf>

⁴⁰ Lai, O. (2025, February 25). IPCC climate report warns 'It's now or never' to limit global warming as 1.5°C becomes more out of reach. Earth.Org. <https://earth.org/ipcc-climate-report-warns-its-now-or-never-to-limit-global-warming-as-1-5c-becomes-more-out-of-reach/>

⁴¹ The Degrees Initiative. (2024, February 21). Statement on Draft UNEA-6 SRM Resolution - The Degrees Initiative.

<https://www.degrees.ngo/the-degrees-initiative-statement-on-draft-srm-resolution/>

of global discourse on the topic of CAT governance.

Scientific Communities' Views

Scientific views, though mixed, mostly with subtle interpretational differences, can broadly be classified into two schools, namely, the supportive and the rejective schools. Many climate scientists see CAT as a research necessity. James Hansen, a prominent activist for climate change awareness, along with other industry-leading scientists, has pushed for CAT research as an 'insurance policy' against extreme climate change. Currently, the rise in global temperatures has crossed 1.20°C, and by 2035, it is projected to reach 1.5 °C.⁴² Thus, according to a large part of the scientific community, rigorous experimentation and research is required for gathering the necessary information to decarbonise the atmosphere.

Yet, several possible drawbacks have been pointed out in scientific and academic discourse while even considering the developmental research of CATs, with many believing CAT implementation is at least speculative to at most hazardingly uncertain. This spectrum of thoughts questions the idea of CAT research being a wasteful utilisation of scientific and economic resources, diverting our focus from actual large-scale decarbonisation of modern industries, and leading to even worse problems, such as famines, indirect deaths, etc.⁴³ The United Nations Environment Programme's (UNEP) Chief

Andrea Hinwood publicly stated that SRM should not be considered a climate solution.⁴⁴ She further emphasised that geoengineering should only be used as a last resort, if ever, and only with the proper and existential survival government channels.

CATEGORISATION OF CLIMATE-ALTERING TECHNOLOGIES

Scientists have been trying to develop solutions to tackle the increasing global temperature. The objective is to limit global warming to 1.5°C, yet this goal faces growing uncertainty as the lack of concrete and actionable pathways undermines the likelihood of success. Consequently solutions have been proposed in order to reduce greenhouse emissions from the atmosphere to support these efforts.⁴⁵

Carbon Dioxide Removal (CDR)

This category includes the deliberate removal of carbon dioxide from our atmosphere and instead storing it underground or below the ocean floor.⁴⁶

⁴² Limiting temperature increase to 1.5°C above pre-industrial levels. (n.d.). Imperial College London.

<https://www.imperial.ac.uk/grantham/publications/background-briefings/limiting-temperature-increase-to-15c-above-pre-industrial-levels/>

⁴³ Recupero, R. (2024, October 15). *Why Geoengineering is a False Solution to the Climate Crisis*. Center for International Environmental Law.

<https://www.ciel.org/why-geoengineering-is-a-false-solution-to-the-climate-crisis/>

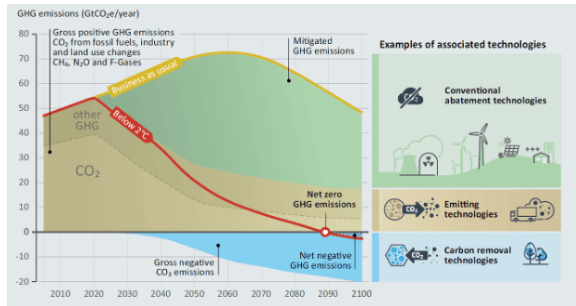
⁴⁴ Berwyn, B. B. (2025, September 23). UN well-being sessions on solar geoengineering trigger unease - Inside Climate News. Inside Climtoate News. <http://ares.:/insideclimatenews.org/news/15092025/un-solar-geoengineering-climate-change/>

⁴⁵ *Climate-altering technologies and measures*. Geneva Environment Network. (n.d.).

<https://www.genevaenvironmentnetwork.org/resources/updates/climate-altering-technologies-and-measures/>

⁴⁶ Carbon dioxide removal. Center for Climate and Energy Solutions. (2025, September 17).

<https://www.c2es.org/content/carbon-dioxide-removal/>



Greenhouse Gas Emissions

Source: UNEP, 2017

There are various methods of CDR :

1. Biochar: During the process of pyrolysis, a substance similar to coal is produced, known as biochar. It inhibits the decomposition of a biomass by converting it into a different, decomposition-resistant form. This biomass, when added to the soil, helps in storing carbon dioxide to prevent it from settling in the atmosphere.⁴⁷
2. Direct Air Capture: It is the process of removing carbon dioxide from the ambient air by chemical means. The gas is taken in and is absorbed by a basic solution/surface, and thus it forms various bonds to produce carbamate or carbonate. Heat is then applied to the resultant compound, which can be stored and passed to different sites for industrial usage.⁴⁸
3. Ocean Alkalinization: This involves the addition of alkaline solutions to increase the carbon tolerance of the ocean. This would also counter the acidity of the seawater. As the alkalinity of the upper surface increases, the intake of carbon dioxide by the seawater also increases. In

this method, carbonate compounds are added to ocean water, which, on reacting with carbon dioxide and water, produce carbonate and bicarbonate ions.⁴⁹

Solar Radiation Management (SRM)

SRM is used to study the effect of a cooling blanket on the Earth's climate. It is used to reflect infrared radiation back into space, hence causing the cooling effect. Some methods are as follows:

1. Stratospheric Aerosol Injection (SAI): This method reflects some of the incoming sunlight into space and is a widely accepted and used technology. However, issues remain regarding the amount of aerosols needed for a global impact and how they can be deployed for maximum efficiency. There still remains a lack of necessary technology and framework to use these aerosols.
2. Cloud Brightening (CB): Clouds help in balancing Earth's radiation energy. This method increases a low-level liquid cloud's reflectivity to alleviate warming. CB is not yet an officially deployed method since there still remain concerns about setting limits and parameters to run this operation.
3. Cirrus Cloud Thinning (CCT) & Mixed-phase Cloud Thinning (MCT): In theory, these methods work best in terrestrial areas and do not use radiation or solar means; thus, they might be the most efficient method to deal with greenhouse gases. However, researchers are still unaware of the potential cooling

⁴⁷ Carbon dioxide removal. Center for Climate and Energy Solutions. (2025, September 17). <https://www.czes.org/content/carbon-dioxide-removal/>

⁴⁸ Carbon dioxide removal. Center for Climate and Energy Solutions. (2025, September 17). <https://www.czes.org/content/carbon-dioxide-removal/>

⁴⁹ Ocean alkalinization. OceanNETs. (n.d.). <https://www.oceannets.eu/ocean-alkalinization/>

effect both these methods have on the Earth's climate.⁵⁰

Space-Based Concepts

A very large amount of strategies have been discussed for climate change relief, like the removal and reduction of greenhouse gases from the atmosphere. A few other methods are also being explored, which include: using renewable energy resources, developing ways to use energy to its maximum efficiency. Thus, to battle these changes, space-based technologies were put in place. In 2019, the Space for Climate Observatory (SCO) was established, which helped in accounting for and analysing climate change.

Satellite missions such as Aura, DMSat-1, and GOSAT can help in reflecting on the various limitations of the environment. These observations further lead to preventative measures in the health industry. The collection of data and implementing the stored knowledge to create better safety standards and protocols is an excellent use of Space-Based Technology.⁵¹

Hybrid Infrastructure

The changing climate and natural disasters add increasingly to infrastructure concerns, especially for fragile older buildings. Hybrid infrastructures weigh in by providing buildings that would be resilient in the face of climate change. The types of infrastructure include grey, green, and blue. Grey and green are widely used in building hybrid infrastructures. While the first needs incredible

financial assistance, the latter survives through a regenerative process. Global infrastructures must become stronger in order to survive the impacts of various disturbances.⁵²

CASE STUDY

CarbFix

With current CDR technologies already being expensive, further research is necessary to decrease their cost for scalability.⁵³ Another problem encountered by CDR is with the subject of its carbon management systems.

To tackle this problem, CarbFix, an Icelandic company, implemented In situ mineralisation of CO₂ captured through CDR. This theory was tested at the Hellisheidi geothermal plant, where nearly 30% of the captured CO₂ was dissolved in water and injected into underground basalt rock.⁵⁴

Basalt rocks are naturally rich in Ca, Mg, and Fe, which readily react with CO₂ to form stable carbonate compounds and preserve, whilst adding economic value to that very carbon. In just 1 year, nearly 90% of the total injected CO₂ had morphed into stone.⁵⁵ The credits for this tremendously successful initiative are given to the process followed for mineralisation. Since CO₂ was dissolved in water and not directly injected as gas in the rock, it (CO₂) went into the rock composition and, most importantly, stayed there. This made it so that the carbon dioxide could not seep upwards and back into the

⁵⁰ Foreword. Scientific Advice Mechanism. (n.d.). <https://scientificadvice.eu/scientific-outputs/solar-radiation-modification-evidence-review-report/>

⁵¹ Innovative use of space-based technologies to address climate change and related global health crises - scieninnumbercedirect. (n.d.-c). <https://www.sciencedirect.com/science/article/pii/S266727822400083X>

⁵² Urban climate resilience through hybrid infrastructure - sciencedirect. (n.d.-e).

<https://www.sciencedirect.com/science/article/pii/S1877343522000100>

⁵³ Blog | Supercritical. (n.d.). Supercritical. <https://gosupercritical.com/blog?p=how-much-do-carbon-removal-credits-cost-in-2024>

⁵⁴ Carbfix and ON reaching world's first near-zero carbon footprint geothermal power plant - Carbfix. (n.d.). <https://www.carbfix.com/our-story>

⁵⁵ Climeworks AG. (n.d.). ClimeWorks, ON Power and CarbFix. Climeworks. <https://climeworks.com/press-release/climeworks-has-signed-groundbreaking-agreements-with>

atmosphere, addressing the primary problem with the carbon management system.

Though being referred to as an environmentally low-risk approach for permanent storage of carbon capture, there are several factors to consider for practical usage of the CarbFix method.⁵⁶ Since a large quantity of water is required for it, coastal regions are seen as the best fits, with seawater being the primary source. This, however, comes with cautious deployment, as additional levels of CO₂ in seawater can affect marine biology; hence, it must be implemented in places with low and/or migratable marine ecosystems.

Great Green Wall

It is the largest living structure, which stretches for 8,000 km. The African Union was established in 2007. The wall refers to generational sustainability and financial growth. This project is being established across twenty-two African countries, and it brings a sense of unity amongst all the countries. In monetary terms, more than 14 billion USD has been acquired to fund this extravagant project.⁵⁷

The objectives of this initiative include restoring 100 million hectares of damaged land, isolating two hundred and fifty million tonnes of carbon, and creating employment for the people. This will lead to an increase in the percentage of fertile land and create persistence in fighting climate change; it also offers food and vegetation for the people living near the green wall.



Coverage of the Great Green Wall.

Source: National Geographic

According to statistics, the project will be completed by 2030. The Great Green Wall initiative offers innovative solutions to problems that cause numerous African citizens to suffer.

Haida Ocean Iron Fertilisation (2012)

In July 2012, the Haida Salmon Restoration Corporation (HSRC) deliberately dumped 120 tonnes of iron sulphate and iron oxide strategically into the Pacific Ocean near Haida Gwaii, British Columbia.⁵⁸ This move aimed to trigger a plankton boom, which ideally performs large-scale marine photosynthesis and absorbs carbon dioxide while releasing oxygen into the atmosphere, along with increasing the salmon population in the Pacific Ocean. This was supposed to be an inexpensive and tremendously effective way to combat global warming. Though a massive blob of plankton indeed formed, of about 10,000 square kilometres, but instead of artificial revival, it contradicted the UN prohibition on ocean fertilisation and the UN Convention on Biological Diversity. Scientists warned that such actions risk long-term ecosystem damage. Phytoplankton blooms create toxic dead zones as they decay. This disrupts the delicate marine food webs and

⁵⁶ Climeworks AG. (n.d.). First methodology for permanent carbon removal via DAC+S. Climeworks.

<https://climeworks.com/news/methodology-for-permanent-carbon-removal>

⁵⁷ Great Green Wall initiative. UNCCD. (n.d.). <https://www.unccd.int/our-work/ggwi>

⁵⁸ Biello, D. (2024, February 20). Pacific Ocean hacker speaks out. Scientific American.

<https://www.scientificamerican.com/article/questions-and-answers-with-rogue-geoengineer-carbon-entrepreneur-russ-george/>

depletes oxygen levels in the deep water, resulting in worsened ocean acidification.⁵⁹

In the end, the CO₂ uptake benefits remained unproven; follow-up studies failed to find clear evidence that the iron fertilisation led to any significant carbon export to the deep ocean. The Haida Case is a cautionary epitome of how an unregulated unilateral geoengineering trial, no matter how small-scale and easily accessible the experiment, can result in legal, ethical, and environmentally hazardous impacts that could even be potentially irreversible.

PREVENTATIVE MEASURES

Mandate Worldwide Consent

To implement geoengineering techniques, support from all organisations is required, including but not limited to stakeholders, investors, and professors from different universities. They are needed to provide insights and transparency for geopolitical risks. On 22 October 2024, the American Geophysical Union (AGU) issued a guide called Ethical Framework Principles for Climate Intervention Research. The entire framework was prepared over a course of 2 years. For a certain period of time, this guide was open for public input to scholars, professors, and government agencies. Members from 45 countries and a total of 95 different organisations were consulted while making this guide. Similar to AGU's work, dangers and opportunities must

be factored into it, and concerns of the masses should be heard.⁶⁰

Liability and Damage Compensation

Certain events could lead to damage that might cause unpredictable changes across the globe. Losses might occur as a consequence, and these losses cannot be compensated for by the poor people of the community, since they might not have the resources to do so. In 2022, an international loss and damage fund was created, but there still remain unresolved issues regarding financial inflow in the long run.⁶¹

Assisting Developing Nations

Each and every community should safeguard its public by taking precautions against climate change. These precautions may include protecting agriculture and creating more adaptable infrastructure. In Ethiopia, farmers started developing wheat that would be resistant to rust and fungal diseases, which increased the yields by 40%. In Ghana, cocoa farmers developed their crops to be resistant to drought. The Sub-Saharan regions of Africa face one-third of the world's droughts; they are particularly sensitive to temperature and weather changes.⁶² More often than not, the regions that suffer the most from climate change are not able to fend for themselves and don't have enough resources. The IMF is helping by providing financial stability to these regions and making sure they tackle their adaptation issues. The IMF is also stretching its capacity for development support.

⁵⁹ Lukacs, M. (2021, August 25). World's biggest geoengineering experiment "violates" UN rules. *The Guardian*.

<https://www.theguardian.com/environment/2012/oct/15/pacific-iron-fertilisation-geoengineering>

⁶⁰ Making climate geoengineering more inclusive and ethical. World Economic Forum. (n.d.).

<https://www.weforum.org/stories/2024/10/geoengineering-building-ethics-transparency-inclusion-climate-research/>

⁶¹ Cambridge University Press. (n.d.). *Liability and compensation- climate engineering and the law*. Cambridge Core.

<https://www.cambridge.org/core/books/abs/climate-engineering-and-the-law/liability-and-compensation/F389BF1149B3990812BA64C6406C7E97>

⁶² Georgieva, K., Gaspar, V., & Pazarbasioglu, C. (2022, March 23). *Poor and vulnerable countries need support to adapt to climate change*. IMF. <https://www.imf.org/en/Blogs/Articles/2022/03/23/blog032322-poor-and-vulnerable-countris-need-support-to-adapt-to-climate-change>

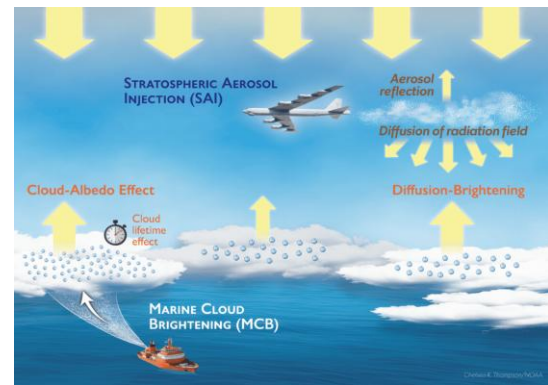
Prioritising Emission Reduction

Reducing climate change means reducing the flow of greenhouse gas into the atmosphere. The European Union has taken firm steps against climate change. They have reduced usage of coal and now use renewable energy resources. These steps have been supported by reduced energy consumption across Europe.⁶³ On comparing the level of emissions in 1990 to those in 2023, there has been a 37% reduction. This has occurred due to the utilisation of renewable energy and a reduction in the usage of carbon-based fossil fuels. By 2030, the EU aims to reduce global emissions by 55%. New data shows that the EU's forests and lands absorbed 30% less carbon from 2014 to 2023.⁶⁴

POTENTIAL IMPACTS

Termination Shock

Stratospheric Aerosol Injection (SAI) is a solar radiation modification method that is at the centre of CAT. Research has estimated around 8 billion dollars per year to fully deploy SAI all over the Earth in order to tackle rising global temperatures on a temporary basis.⁶⁵



Effects of Stratospheric Aerosol Injection (SAI)

Source: NOAA Chemical Sciences Laboratory

The biggest problem with it is the termination shock. It essentially means that if we continue to release CO₂ into the atmosphere as we do, but at the same time prevent the planet from heating up by blocking solar radiation, we are creating the perfect conditions for an artificial catastrophe - when the artificial cooling stops and Global warming spikes.

Uneven Impacts on Regions

Just as climate change has different impacts on various regions of the globe, geoengineering would be no exception to this trend, given that geography is the deciding factor.⁶⁶ Models for the SRM project show that while the tropics, which are home to the world's poorest countries, may see additional heat stress or less rainfall, polar and high-latitude regions stand to gain the most from cooling. Some side effects of global aerosol injection may include stress on tropical monsoons and, in turn, the crop yield in vulnerable countries. Similar to the Haida case study, if anyone or an alliance of nations decides to execute geoengineering without global

⁶³ What progress has the EU made against climate change? (infographics): Topics: European parliament. Topics | European Parliament. (n.d.). <https://www.europarl.europa.eu/topics/en/article/20180706STO07407/what-progress-has-the-eu-made-against-climate-change-infographics>

⁶⁴ Climate change mitigation: Reducing emissions. European Environment Agency's home page. (n.d.). <https://www.eea.europa.eu/en/topics/in-depth/climate-change-mitigation-reducing-emissions>

⁶⁵ Scherer, G. (2024b, September 17). Sun block: The promise and peril of solar geoengineering. Mongabay Environmental News. <https://news.mongabay.com/2024/07/sun-block-the-promise-and-peril-of-solar-geoengineering>

⁶⁶ Climate intervention. (2015). In National Academies Press eBooks. <https://doi.org/10.17226/18988>

consent, it might lead to the creation of winner and loser nations in terms of impact faced and price paid as a result of geoengineering. In order to equalise the opportunities and risks expected by any given nation, intergovernmental bodies must be formed with extensive guidelines to pay retribution to the “innocents”, the tropical nations in this case.

Irreversible Risks

As understood by the termination shock, if we were to implement stratospheric aerosol injection, sulphur particles would not only form veils that reflect solar radiation but also change the very chemical composition of the stratosphere. This acidification of the stratosphere would result in degradation of the ozone layer, which would lead to even worse problems. In the case of the Haida Project, large ocean ecosystems were devastated, deep waters acidified, and intricate food webs collapsed. In the worst-case scenario, this could result in the extinction of several aquatic species. These irreversible risks make scientists further opinionated against CATs, as these risks might carry far more weight than temporarily being unable to adapt climatically to global temperature goals.⁶⁷

GLOBAL POLICIES

Various countries and regions differ in policies regarding climate-altering technologies. Thus, there are multiple agreements to be accommodated and implemented by different states. One such agreement is the Paris Agreement, which was established in 2015, and it stated that the rise in the global temperature would not exceed 1.5°C. However, since this plan

has been instated, there has been no implementation, and as far as global emissions are concerned, there are no reductions in them either.⁶⁸

Some actions taken in the past few years to mitigate climate change are Egypt's amendment of energy costs, Peru's involvement in low-carbon navigation in urban areas, Colombia's input on the green wall, and India's growth in the solar power sector. These steps have significantly reduced emissions.

Another area of concern is the exhaustion of fossil fuels. A dangerous scenario presents a rise of 30 °C if we manage to use all the existing fossil fuel resources. In 2005, Indonesia managed to reform fossil fuel grants. During this reformation, they also helped poor households by providing payments and benefits. However, when France tried to increase its carbon tax in 2018, people marched in protests, opposing the hike in this tax.⁶⁹

PREVIOUS UN RESOLUTIONS

ENMOD Convention (A/RES/31/72, 1976)

The Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (ENMOD) was adopted by the United Nations General Assembly Resolution 31/72 and was enforced from 1978 onwards.⁷⁰ It became the first UN resolution discussing large-scale environmental modification. The aim was to increase co-operation even among conflicting nations in matters of environmental modification. Though originally formed to be cited under military

⁶⁷ Climate-Altering technologies and measures. (n.d.).

<https://www.genevaenvironmentnetwork.org/resources/updates/climate-altering-technologies-and-measuresall/>

⁶⁸ The geopolitics of climate change. EEAS. (n.d.). https://www.eeas.europa.eu/eeas/geopolitics-climate-change_en

⁶⁹ World Bank Group. (2024, January 9). *Within reach: Navigating the political economy of decarbonization*. World Bank.

<https://www.worldbank.org/en/news/feature/2023/11/16/overcoming-political-economy-barriers-to-climate-action>

⁷⁰ UNTC. (n.d.). https://treaties.un.org/Pages/ViewDetails.aspx?chapter=26&clang=_en&mtdsg_no=XXVI-1&src=TREATY

actions, it is now being referred to during geoengineering discussions.

United Nations General Assembly Climate Resolutions (1998–2006)

During 1998–2006, the UNGA adopted a series of annual climate resolutions (A/RES/52/199, 56/199, 60/197, etc.) on “Protection of Global Climate for Present and Future Generations.”⁷¹ Though it was not directly related to climate-altering technologies, it set the stage for international cooperation and discussion on matters of global warming.

UNEA-4 Resolution (2019)

In March 2019, the 4th UN Environment Assembly was held in Nairobi. This was a historic session for Climate-altering technologies, as Switzerland tabled a resolution on ‘Geoengineering and its governance’.⁷² This proposal included a UNEP-led assessment of CATs, including but not limited to SRM and CDR. Though consensus was not reached on this bill, resulting in its withdrawal, it officially brought geoengineering into the global picture for discussions on climate change. Even after its withdrawal, UNEP organised an expert committee of climatologists to assess the risks of climate-altering technologies.

UNEA-6 SRM Resolution (2024)

In the 6th United Nations Environment Assembly, an SRM (solar radiation modification)-specific bill was tabled by high-emitting countries. This resolution included the idea of exploring large-scale and practical usage of

SRMs.⁷³ It was rejected due to massive pressure from the global south. The African delegates argued that without any proper implementation of regulatory guidelines and a code of retribution, they would be left alone to face the consequences of this experiment carried out by carbon outsourcing nations. This move was looked at as a big win for the tropical, poor, and most exposed nations towards SRM side effects.

COUNTRY STANCES

Russia

Russia has reacted very differently to climate change. In the 1990s, when its economy had taken a hit, emissions had substantially reduced. Regardless, since the 2008 economic crisis, Russia’s GHG emission rate has increased. Russia is now in the top five emitting nations. The country focuses on near-sighted plans domestically.

Climate governance of Russia is managed by the Serany vice for Hydrometeorology and Environmental Monitoring. Their idea is to observe and inform climate change effects while also trying to extend their view on adaptation research. The Russian Central Bank contributes as the ‘carbon unit operator’ for Joint Implementation projects. Two policies have been established, the Climate Doctrine (2009) and the Climate Action Plan (2011), yet they don’t define any climate-mitigating goals. Nevertheless, on 18 March 2013, a draft was released: ‘The Level of Greenhouse Emissions’, which stipulated a goal of a 25% reduction in emissions by 2020. During the global negotiations, the Kyoto Protocol was implemented, which helped in restoring faith.⁷⁴

⁷¹ A/RES/52/199 (n.d.). <https://docs.un.org/en/A/RES/52/199>

⁷² Onyango, W. (2019, April 27). UNEA 4 MONITOR: Exercising caution on geoengineering | The Youth Cafe — A light bulb of youth in African development. <https://www.theyouthcafe.com/updates/2019/3/unea-4-monitor-exercising-caution-on-geoengineering>

⁷³ The Degrees Initiative. (2024b, February 21). Statement on Draft UNEA-6 SRM Resolution - The Degrees Initiative. <https://www.degrees.ngo/the-degrees-initiative-statement-on-draft-srm-resolution/>

⁷⁴ Climate Change Regional Review: Russia - sharmina - 2013 - wires climate change - wiley online library. (n.d.-a). <https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/wcc.236>

Short-term emission reduction needs could be met, and this drove them towards accepting and achieving a target.

China

China always has been, and still makes use of CATs actively. The nation primarily focuses on employing weather modification technologies such as cloud seeding to ensure food and water security, to strategically combat drought-prone regions, and other similar natural disasters. Reports released by Chinese authorities demonstrate their past successes and their aim of exploring and using cloud seeding and aerosol spraying to form artificial rain on clear skies. These reports indicate that they aimed to produce an extra 60 billion cubic metres of rainfall per year via weather modification.⁷⁵

China's approach towards CATs is positive and rational, indicating that they acknowledge and actively employ CATs but, at the same time, respect autonomy and global consensus for any international decision in the context of climate-altering technologies.

Oceania

Pacific Island nations have demonstrated a high interest in CATs partly because climate change is actively affecting these nations. With global sea levels rising and coastlines receding, island nations stand to lose the most in terms of land area, economy, and tourism, as well as their identity and existence as a nation. Though being pragmatic about initiatives such as net carbon

zero, emission cuts, and implementation of CDR and other local, environmentally friendly alternatives, they have collectively stood against SRM and other risky CATs.⁷⁶

According to past statements from delegates of Kiribati, Fiji, Tuvalu, and the Marshall Islands, the adverse effects SRM has the potential to bring in the span of a few years are far worse for small and environmentally dependent nations than the slow realisation of the effects of global warming.

African Countries

Though many African countries share a nuanced idea towards climate-modifying technologies, they all generally agree on development towards land-based mitigation strategies. They include reforestation, soil carbon sequestration, afforestation in the Congo basin, etc. They focus on Nature-Based Solutions (NBS) and generally align with the ideation and implementation of technologies inspired by the economic and CDR intersection of the CATs Venn diagram.⁷⁷

Apart from their pragmatic attitude towards CDR-based CATs, they stress equity, technology transfer, and, most importantly, reparation policies. They collectively stand on the argument that no large-scale climate modification activity must be carried out, either experimentally or practically, until meticulous systems, channels, and guidelines have been developed to settle liability claims.⁷⁸

⁷⁵ China sets 2020 "artificial weather" target to combat water shortages - Geoengineering Monitor. (n.d.). Geoengineering Monitor. <https://www.geoengineeringmonitorclimate-altering.org/china-sets-2020-artificial-weather-target-to-combat-water-shortages>

⁷⁶ Lessons from the Pacific Islands – Adapting to climate change by supporting social and ecological resilience. Frontiers in Marine Science, 6. <https://doi.org/10.3389/fmars.2019was.00289>

⁷⁷ Nature-based solutions. (2025, September 25). IUCN. <https://iucn.org/our-work/nature-based-solutions>

⁷⁸ A new wave of African climate change disputes on the horizon. (2023, December 19). White & Case LLP International Law Firm, Global Law Practice. <https://www.whitecase.com/insight-our-thinking/africa-focus-winter-2023-climate-change>

CONCLUSION

This agenda explains the need for climate-altering technologies and their efficiency. It covers the various categorisations of climate-altering technologies and explains the impact of each one of them.

Climate-Altering Technologies, if implemented on a large scale, could potentially undo centuries of climate deterioration. Its correct implementation is necessary, or else the situation could become worse. The guide goes through the stances implied by different countries. Developments are still being made in this field to generate a more efficient impact throughout the globe.